```
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'Project 9
'04/03/2024

' {$STAMP BS2}
' {$PBASIC 2.5}

time      VAR Word
row       VAR Nib
column    VAR Nib
keypad    VAR Word
keypadOld VAR Word
temp      VAR Nib
constant  VAR Word
C1        VAR Byte
C2        VAR Byte
C3        VAR Byte
C4        VAR Byte
C5        VAR Word
password1 VAR Nib(6)
password2 VAR Nib(6)
char      VAR Byte
value     VAR Byte
tone      VAR Byte

constant = 0
C2 = 0
C3 = 0
C4 = 0
C5 = 0

PAUSE 100
SEROUT 14, 84, [22, 12, 17]
LOW 13
LOW 12
LOW 11
GOTO Start

Start:
    C1 = 3
    SEROUT 14, 84, [128, "System ready        ", 148, "Enter Passcode:"]
    DO
    GOSUB ReadKeypad
    PAUSE 100
    IF keypad THEN
    IF keypad = keypadOld THEN
    PAUSE 50
    ELSE
    GOSUB Display
    time = 0
    ENDIF
    ELSE
    time = time + 1
    IF time >= 25 THEN
    SEROUT 14, 84, [12, 128, "Timed out wrong code"]
    PAUSE 2000
    C2 = 0
    C3 = 0
    C4 = 0
    time = 0
    GOTO Start
    ENDIF
    ENDIF
    keypadOld = keypad
    LOOP UNTIL (C2 = 7)
    GOTO Invalid
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    RETURN

ReadKeypad:
    keypad = 0
    OUTL = %00000000
    DIRL = %00000000
    FOR row = 0 TO 3
    DIRB = %1111
    OUTB = %0000
    OUTA = 1 << row
    DIRA = 1 << row
    temp = 0
    FOR column = 0 TO 3
    INPUT (column + 4)
    temp = temp | (INB & (1 << column))
    NEXT
    keypad = keypad << 4 | (Temp REV 4)
    NEXT
    RETURN

Display:
    FOR temp = 0 TO 15
    IF (keypad & (1 << temp)) THEN
    LOOKUP temp, ["D","#","0","*","C","9","8","7","B","6","5","4","A","3","2","1"], char
    LOOKUP temp, [13,15,0,14,12,9,8,7,11,6,5,4,10,3,2,1], value
    LOOKUP temp, [224,223,222,221,220,230,229,228,227,226,225,224,223,222,221,220], tone
    SEROUT 14, 84, [188 + C2, char, tone]
    IF value < 14 THEN
    IF (C1 = 3) THEN
    password1(C2) = value
    ELSE
    password2(C2) = value
    ENDIF
    C2 = C2 + 1
    IF value <= 9 THEN
    C3 = 1
    ELSE
    C4 = 1
    ENDIF
    ELSEIF value = 14 THEN
    IF (C1 = 3) THEN
    C2 = C2 + 1
    ELSEIF (C1 = 1 OR C1 = 4 AND C2 = 6) THEN
    GOTO P_Check
    ELSEIF (C1 = 2 AND C2 = 6) THEN
    GOSUB C_Check
    C2 = 0
    GOTO Disarm
    ELSE
    password1(C2) = value
    C2 = C2 + 1
    ENDIF
    ELSEIF value = 15 THEN
    IF (C1 = 3) THEN
    IF (C2 = 6 AND C3 = 1 AND C4 = 1) THEN
    GOTO Disarm
    ELSE
    GOTO Invalid
    ENDIF
    ELSEIF (C1 = 2 AND C2 = 6) THEN
    GOTO P_Check
    ELSEIF (C1 = 1 OR C1 = 4 AND C2 = 6) THEN
    GOSUB C_Check
    C2 = 0
    IF C1 = 1 THEN
    GOTO Armed
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    ELSE
    GOTO Caution
    ENDIF
    ELSE
    password2(C2) = value
    C2 = C2 + 1
    ENDIF
    ENDIF
    PAUSE 7
    EXIT
    ENDIF
    NEXT
    RETURN

Invalid:
    SEROUT 14, 84, [12]
    SEROUT 14, 84, [128, "Invalid!"]
    PAUSE 3000
    C3 = 0
    C4 = 0
    C2 = 0
    GOTO Start
    RETURN

Armed:
    C2 = 0
    SEROUT 14, 84, [12, 17]
    SEROUT 14, 84, [132, "System Armed", 151, "Enter Password", 173, "To Disarm"]
    C1 = 1
    SEROUT 14, 84, [218, 223]
    SEROUT 14, 84, [218, 228]
    DO
    HIGH 13
    PAUSE 50
    LOW 13
    PAUSE 50
    GOSUB ReadKeypad
    PAUSE 100
    IF keypad THEN
    IF keypad = keypadOld THEN
    PAUSE 50
    ELSE
    GOSUB Display
    time = 0
    ENDIF
    ELSE
    time = time + 1
    IF time >= 25 THEN
    SEROUT 14, 84, [12, 128, "Timed out wrong code"]
    PAUSE 2000
    C2 = 0
    C3 = 0
    C4 = 0
    time = 0
    GOTO Armed
    ENDIF
    ENDIF
    keypadOld = keypad
    IF(C2 = 7) THEN
    GOSUB C_Check
    GOTO Armed
    ENDIF
    IF (C2 > 0) THEN
    ENDIF
    IF (IN15 = 1) THEN
    GOTO Caution
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    ENDIF
    LOOP
    RETURN

Caution:
    C1 = 4
    C2 = 0
    DO
    IF C5 = 0 THEN
    SEROUT 14, 84, [129, " CAUTION Intruder!  ", 151, "Enter Password", 173, "To Disarm"]
    SEROUT 14, 84, [218, 219, 230]
    SEROUT 14, 84, [218, 219, 229]
    SEROUT 14, 84, [218, 219, 228]
    HIGH 12
    PAUSE 10
    ELSEIF (C5 = 2) THEN
    SEROUT 14, 84, [21]
    PAUSE 100
    SEROUT 14,84, [22]
    PAUSE 100
    LOW 12
    PAUSE 10
    ENDIF
    C5 = C5 + 1
    IF (C5 = 4) THEN
    C5 = 0
    ENDIF
    GOSUB ReadKeypad
    PAUSE 100
    IF keypad THEN
    IF keypad = keypadOld THEN
    PAUSE 50
    ELSE
    GOSUB Display
    time = 0
    ENDIF
    ELSE
    time = time + 1
    IF time >= 25 THEN
    SEROUT 14, 84, [12, 129, "    Timed out", 151, "  wrong code"]
    PAUSE 2000
    C2 = 0
    C3 = 0
    C4 = 0
    time = 0
    GOTO Caution
    ENDIF
    ENDIF
    keypadOld = keypad
    IF(C2 = 7) THEN
    GOSUB C_Check
    C2 = 0
    ENDIF
    LOOP
    RETURN

Disarm:
    C2 = 0
    C1 = 2
    LOW 12
    SEROUT 14, 84, [12]
    SEROUT 14, 84, [130, "System is Ready", 151, "Enter Password", 174, "To Arm"]
    DO
    HIGH 11
    PAUSE 50
    LOW 11
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```
    PAUSE 50
    GOSUB ReadKeypad
    PAUSE 100
    IF keypad THEN
    IF keypad = keypadOld THEN
    PAUSE 50
    ELSE
    GOSUB Display
    time = 0
    ENDIF
    ELSE
    time = time + 1
    IF time >= 25 THEN
    SEROUT 14, 84, [12, 128, "Timed out wrong code"]
    PAUSE 2000
    C2 = 0
    C3 = 0
    C4 = 0
    time = 0
    GOTO Disarm
    ENDIF
    ENDIF
    keypadOld = keypad
    IF(C2 = 7) THEN
    GOSUB C_Check
    GOTO Disarm
    ENDIF
    LOOP
    RETURN

P_Check:
    LOW 12
    FOR C5 = 0 TO 5
    IF (password2(C5) = password1(C5)) THEN
    PAUSE 100
    ELSE
    GOSUB C_Check
    IF C1 = 4 THEN
    C2 = 0
    GOTO Caution
    ELSEIF C1 = 1 THEN
    GOTO Armed
    ELSE
    GOTO Disarm
    ENDIF
    ENDIF
    NEXT
    C5 = 0
    SEROUT 14, 84, [12]
    SEROUT 14, 84, [128, "Password Correct"]
    C2 = 0
    PAUSE 2000
    IF C1 = 1 OR C1 = 4 THEN
    GOTO Disarm
    ELSE
    GOTO Armed
    ENDIF
    RETURN

C_Check:
    SEROUT 14, 84, [12]
    SEROUT 14, 84, [128, " Password Incorrect"]
    PAUSE 200
    SEROUT 14, 84, [218, 227]
    PAUSE 100
    SEROUT 14, 84, [218, 226]
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PAUSE 100
SEROUT 14, 84, [218, 225]
PAUSE 1000
RETURN
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